# The exact maximum in Erdős problem 272 for sufficiently large ground sets

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**Abstract**

Let $v(N)$ be the maximum cardinality of a family of distinct subsets of $\{1,\dots,N\}$ whose intersections of distinct members are nonempty arithmetic progressions. We give a computer-assisted proof that

$$v(N) = \binom{N}{2} + 1 + \left\lfloor \frac{N-1}{4} \right\rfloor \qquad (N \geq 200000).$$

The lower bound is Szabó's construction. For the upper bound we delete large non-progressions, obtain a common point for the remaining nonprogressions, and force all members avoiding that point to be long intervals. Their common interval then pays for the deleted members. We give the explicit numerical estimates and the precise interfaces of the formally proved counting lemmas. For the finite-window argument we combine a source from a longest AP with a totient source, use a zero-cost initial potential layer, and perform two symbolic recovery preparations. The accompanying Lean 4.35.0 development supplies the complete structural proofs and finite certificates and forms part of the proof. The remaining finite range below 200000 is open in this development.

## 1 Statement and scope

Write $[N] = \{1,\dots,N\}$. An arithmetic progression (AP) means a nonempty finite set $\{a, a+d, \dots, a+kd\}$ with $d \geq 1$ and $k \geq 0$. Thus singletons and pairs are APs. A family $\mathcal{F} \subseteq 2^{[N]}$ is *legal* if $S \cap T$ is an AP whenever $S, T \in \mathcal{F}$ and $S \neq T$. The members themselves need not be APs. Set

$$B(N) = \binom{N}{2} + 1 + \left\lfloor \frac{N-1}{4} \right\rfloor, \qquad v(N) = \max_{\mathcal{F} \text{ legal}} |\mathcal{F}|.$$

The maximum exists because its domain is finite.

**Theorem 1.** *For every integer $N \geq 200000$, $v(N) = B(N)$. For every $N \geq 1$, there is a legal family of size $B(N)$.*

This is a partial answer to Erdős problem 272 and the corresponding Justin Sun Prize problem JSP-000233. Szabó established the leading term and the construction giving $B(N)$ [1]. Yang formulated the exact formula and proved the common-point bound using private pairs and a staircase inequality [2]. Our formal development also follows the non-common-point structural route of Kohlmeyer and Kruer [3]. We do not claim priority for those constructions or methods. We prove existence of an optimal common-point family, not that every optimal family has a common point.

*Project and complete proof sources: https://github.com/lixiang90/Erdos272/tree/dev.

The theorem is *computer-assisted*: the finite-set counting lemmas specified below have complete proofs in the companion Lean source, which forms an electronic supplement to this paper. The paper supplies their mathematical interfaces and the full numerical and logical assembly. It does not replace those source proofs by an assertion that a numerical search succeeded. Section 6 gives direct paths and reproduction commands.

## 2 The matching construction

Let $c = \lceil N/2 \rceil$ and $k = \lfloor (N-1)/4 \rfloor$. Start with every subset of $[N]$ of size at most three containing $c$. For each $1 \le d \le k$, remove the two triples

$$\{c-2d, c, c+d\}, \qquad \{c-d, c, c+2d\},$$

and add the five-term progression $P_d = \{c-2d, c-d, c, c+d, c+2d\}$ and its two four-term end deletions.

**Proposition 2** (Szabó's construction)**.** *For every integer $N \ge 1$, the resulting family is legal and has cardinality $B(N)$.*

*Proof.* Since $2k \le \min(c-1, N-c)$, all its points lie in $[N]$. Every member contains $c$. Intersections involving a member of size at most two, or two distinct triples, therefore have size one or two. Two APs have an AP as their nonempty intersection: their common points form a residue class modulo the least common multiple of their steps, restricted to their common span.

It remains to intersect a retained triple with an added AP. Only a three-point intersection needs checking. Such a triple lies in some $P_d$ and contains $c$. After translating and scaling, the six possible triples through zero in $\{-2,-1,0,1,2\}$ consist of four APs and the two deleted triples $\{-2,0,1\}$ and $\{-1,0,2\}$. Thus every required intersection is an AP.

The ordered gaps $(2d, d)$ and $(d, 2d)$ distinguish the deleted triples. The step and endpoints distinguish the added APs. Consequently exactly $2k$ triples are removed and $3k$ larger members are added, giving

$$N + \binom{N-1}{2} - 2k + 3k = \binom{N}{2} + 1 + k.$$

This also covers $k = 0$. □

## 3 The structural counting certificates

A member is *crooked* if it is not an AP. For a legal family $\mathcal{F}$, define

$$\mathcal{D}(\mathcal{F}) = \{S \in \mathcal{F} : S \text{ is crooked and } 28|S| > N\}, \qquad \mathcal{G}(\mathcal{F}) = \mathcal{F} \setminus \mathcal{D}(\mathcal{F}).$$

Every AP member is retained. Write $H_N = \sum_{i=1}^{N} 1/i$ and $J = \lfloor \log_2 N \rfloor + 1$. All intervals below are integer intervals, with both endpoints included.

We use the following formally proved structural facts. Their hypotheses refer to the actual family and its subsets. In particular they do not assume an upper bound on an abstract collection of witnesses.

**Lemma 3** (Preparation and endpoint deficit)**.** *Suppose $N \ge 2^{49}$, $\mathcal{F}$ is legal, and $|\mathcal{F}| > B(N)$. Either leave $\mathcal{F}$ unchanged or reflect every member by $x \mapsto N+1-x$, obtaining a legal family $\mathcal{H}$ with $|\mathcal{H}| = |\mathcal{F}|$. There are integers $1 \le c < L \le R \le N$ with the following properties. Put $\mathcal{D} = \mathcal{D}(\mathcal{H})$, $\mathcal{G} = \mathcal{G}(\mathcal{H})$, and $\mathcal{A} = \{S \in \mathcal{G} : c \notin S\}$.*

1. *Every crooked member of $\mathcal{G}$ contains $c$, and $\mathcal{A} \neq \varnothing$.*

2. *Every $S \in \mathcal{A}$ has $\min S \leq L$ and $\max S \geq R$.*

3. *With $t = L + N - R$, there is a nonnegative witness count $E$ such that*

$$|\mathcal{D}| \leq 3137N, \qquad E \leq 6NH_N^3, \tag{1}$$

$$|\mathcal{H}| \leq \frac{N^2}{2} - \frac{t^2}{102} + |\mathcal{D}| + E + 40NH_N^2 + 6NH_N + 24N + (L - c) + 3. \tag{2}$$

Here and below the statements of the source lemmas are stronger than the specializations displayed. For Lemma 3, the exact source is `Counting/MiddleDeficit52.lean`, declaration `actual_original_prepared_endpoint102`; deletion and the exceptional count are supplied by `Core/DenseSharper41.lean` and `Arithmetic/DivisorMoment45.lean`. All source paths in the paper are relative to `formal/Erdos272/`.

We describe the counting mechanism to clarify what is certified. A crooked set containing a root $c$ has a pair $\{x, y\}$ for which the smallest AP containing $c, x, y$ is not contained in the set. Such a pair cannot belong to another member through $c$, since their legal intersection would contain that AP. These private pairs, and the pairs covered by AP members, are counted in the same pair universe. Deleting large crooked members first bounds their cost by $(4 \cdot 28^2 + 1)N = 3137N$. The remaining crooked members have a common point by the empty-kernel counting bound. If the retained family itself has a common point, the common-point bound and the omitter estimate already exclude a counterexample. Otherwise an avoiding AP supplies a one-sided orientation, and endpoint extrema supply $L, R$.

The endpoint estimate counts primitive pairs in the middle region; coprimality, floor errors, and exceptional witnesses are retained in (2). The coefficient $1/102$ is deliberately conservative. These combinatorial arguments, including the common-point bound and its finite staircase certificate and unbounded tail, are proved in the import closure of the cited module. No kernel conjecture is assumed.

**Lemma 4** (Excluding noninterval avoiders)**.** *Let $\mathcal{G}$ be legal, let $c \in [N]$, and suppose every crooked member of $\mathcal{G}$ contains $c$. Suppose $c \leq L \leq N$, and each member avoiding $c$ contains a point at most $L$. If an avoiding member is not an integer interval, then*

$$|\mathcal{G}| \leq \frac{3N^2}{8} + \frac{5}{3}LN + 2NH_N + 20N + 3 + 6NH_N^3. \tag{3}$$

*In particular, if $N \geq 6$, $N < 2^J$, $J \geq 2$, $32L \leq N$, and $16384J^3 \leq N$, then $|\mathcal{G}| \leq 11N^2/24$.*

The exact finite-set counting proof is `Global/NonInterval53.lean`, declarations `actual_noninterval_family_count` and `actual_noninterval_family_gap`. An avoiding member is an AP. If it is not an interval, its step is at least two and its size is at most $N/2 + 1$. The unordered pairs meeting it number at most $3N^2/8 + N$. Private witnesses for crooked members inject into these pairs, except for the separately bounded exceptional members. Counting the AP members with a low endpoint gives (3). For the stated consequence, use $H_N \leq J$ and $J \geq 2$. The last four terms in (3) are at most $31NJ^3 \leq 31N^2/16384$. Hence

$$\frac{3}{8} + \frac{5}{96} + \frac{31}{16384} < \frac{11}{24}.$$

The elementary inequality $H_N \leq J$ follows by partitioning $[N]$ into the dyadic blocks $\{1\}, [2, 3], [4, 7], \ldots$; each contributes at most one.

**Lemma 5** (Interval count and restoration)**.** *Let $N \geq 29^3$ and $1 \leq c < L \leq R \leq N$, and put $t = L + N - R$. Let $\mathcal{H}$ be legal, $\mathcal{D} = \mathcal{D}(\mathcal{H})$, and $\mathcal{G} = \mathcal{G}(\mathcal{H})$. Suppose the members of $\mathcal{G}$ avoiding $c$ form a nonempty family of intervals $[a, b]$ satisfying $c < a \leq L \leq R \leq b \leq N$. If*

$$6tJ \leq N, \qquad 812t \leq N, \tag{4}$$

*then*

$$|\mathcal{G}| \leq \binom{N}{2} + 1, \qquad |\mathcal{D}| \leq 131(t-1) \leq \left\lfloor \frac{N-1}{4} \right\rfloor. \tag{5}$$

*Consequently $|\mathcal{H}| \leq B(N)$.*

The first bound is `Retention/Terminal36.lean`, declaration `actual_native_interval_terminal`. The second is `Retention/JointFringe44.lean`, declaration `actual_dense_fringe_bound`, with parameters $r = 28$, $p = 29$. Their assembly is `Global/GlobalUpper53.lean`, declaration `actual_deleted_long_core_restore28`.

Here is why the two counts fit together. An avoiding interval contains $[L, R]$, so this core lies in an actual AP member of the original family. Its exterior in $[N]$ has cardinality $t - 1$. A deleted member has more than $N/28$ points. The inequality $28 \cdot 29t \leq N$ forces a long trace in the core. Bounded-step trace anchors and points outside the core give the joint bound

$$\mathrm{JointCount}(29)(t-1) = \left(29 + \sum_{d=1}^{28} \left\lfloor \frac{29}{d} \right\rfloor\right)(t-1) = 131(t-1).$$

The aligned and off-residue cases are both included in this count. For the final payment, $4 \cdot 131t = 524t \leq 812t \leq N$ implies $4 \cdot 131(t-1) \leq N - 1$. Since $\mathcal{D}$ and $\mathcal{G}$ partition the original family, their bounds add to exactly $B(N)$.

## 4 The explicit large-order argument

**Proposition 6** (Large-order tail)**.** *For every integer $N \geq 2^{49}$, $v(N) = B(N)$.*

We prove Proposition 6 from the certified counting lemmas. Every numerical estimate in this section is elementary.

**Lemma 7** (Uniform logarithmic payments)**.** *If $N \geq 2^{49}$ and $J = \lfloor \log_2 N \rfloor + 1$, then*

$$J \geq 50, \qquad 470771616J^3 \leq N, \qquad 25704J^5 \leq N. \tag{6}$$

*Proof.* Put $k = J - 1 \geq 49$. At $k = 49$, direct integer arithmetic gives

$$470771616 \cdot 50^3 = 58846452000000 < 2^{49}, \qquad 25704 \cdot 50^5 = 8032500000000 < 2^{49}.$$

For $k \geq 49$, $(k+2)/(k+1) \leq 51/50$, and $(51/50)^5 < 2$. Thus both $(k+1)^3/2^k$ and $(k+1)^5/2^k$ decrease by induction. The desired bounds follow from $2^k \leq N$. □

*Proof of Proposition 6.* Proposition 2 proves the lower bound. Suppose for a contradiction that $N \geq 2^{49}$ and $|\mathcal{F}| > B(N)$ for a legal family. Apply Lemma 3 and use its notation. Since $H_N \leq J$, $L - c \leq N$, and $N \geq 1$, (1)–(2), together with $|\mathcal{H}| \geq \binom{N}{2}$, imply

$$t^2 \leq 102N(6J^3 + 40J^2 + 6J + 3166) \leq 714NJ^3. \tag{7}$$

For the last inequality, $J \geq 50$ and the identity

$$J^3 - 40J^2 - 6J - 3166 = (J - 50)(J^2 + 10J + 494) + 21534 \geq 0$$

suffice. The slack in 3166 also covers the $N/2$ difference between $N^2/2$ and $\binom{N}{2}$, and the constant 3 in (2).

As $812^2 \cdot 714 = 470771616$ and $36 \cdot 714 = 25704$, Lemma 7 and (7) give

$$(812t)^2 \le N^2, \qquad (6tJ)^2 \le N^2.$$

All quantities are nonnegative, so $812t \le N$ and $6tJ \le N$. Also $L \le t$, whence $32L \le N$, and $16384J^3 \le 470771616J^3 \le N$.

If an avoiding member of $\mathcal{G}$ were not an interval, Lemma 4 would give $|\mathcal{G}| \le 11N^2/24$. On the other hand, $|\mathcal{H}| > B(N)$ and $|\mathcal{D}| \le 3137N$ imply

$$\frac{N^2}{2} \le |\mathcal{G}| + 3138N.$$

These inequalities would force $N \le 24 \cdot 3138 = 75312$, contrary to $N \ge 2^{49}$. Therefore every avoiding member is an integer interval $[a, b]$. The endpoint conditions in Lemma 3 give $a \le L \le R \le b$. Since $c < L$ and $c \notin [a, b]$, necessarily $c < a$.

All hypotheses of Lemma 5 now hold, for the same family, root, and endpoints. It follows that $|\mathcal{F}| = |\mathcal{H}| \le B(N)$, a contradiction. □

## 5 The finite-window certificates

The finite part of the argument uses deletion parameter 12. Its source and recovery lemmas apply to the same actual legal family throughout. We give the certificate interfaces and the choices at the lower end of the range. All their finite proof data are included in the source.

For a member $S$, let $s = |S|$, let $a$ be the largest cardinality of an AP contained in $S$, and put $e = s - a$. Let $\mathrm{Pool}(S)$ be the three-element subsets of $S$ whose AP hull is not contained in $S$. The longest-AP source gives

$$|\,\mathrm{Pool}(S)| \ge eU(a), \qquad U(a) = \binom{a}{2} - \sum_{\substack{p<a \\ p \text{ prime}}} \binom{\lceil a/p \rceil}{2}.$$

It counts pairs in the AP and one point outside it. The sieve removes pairs whose hull can remain in $S$. An unused prime class pays for an exceptional maximal run, so the estimate is uniform in the actual step denominator. The exact source is `Core/N200000/CoreUnionSource200000.lean`.

For $a \le 2^k$, prime reciprocal bounds imply

$$U(a) \ge \frac{a^2}{4} - \frac{k+1}{2}a.$$

The certificate uses $k = 8$ up to 256 and $k = 18$ above 256, with a uniform tail price 14070 from 257. The monotonicity used here is that of the displayed quadratic on the relevant domain.

The totient source supplies a second lower bound, denoted $\mathrm{Floor}(T, D, H)$, asserting that every relevant member with $s \ge T$ and $e \ge D$ has at least $H$ pool triples. To pay a target $H$ when only $e \ge E$ is known, the mixed certificate checks

$$\mathrm{Floor}(T, D, H), \qquad H \le Eu, \qquad u \le U(a) \quad (T - D + 1 \le a \le 2^{18}).$$

If $e \ge D$ the first bound applies. If $e < D$, then $a \ge T - D + 1$ and the longest-AP source gives $eU(a) \ge Eu \ge H$. This implication is `Core/N200000/MixedCore200000.lean`, declaration `actual_mixed_source`. There are 150 source plans in the three windows below.

| $N$ range | Member floor | Free height | Final height | Paid layers |
|---|---|---|---|---|
| $[200000, 210000]$ | 134 | 3402 | 175390 | 7 |
| $[210001, 220000]$ | 137 | 3520 | 184410 | 7 |
| $[220001, 233810]$ | 140 | 3700 | 193464 | 7 |

A crooked member has $e \geq 1$, so the mixed source already pays the free height. The family with pool strictly below that height is empty. Its initial potential-layer cardinality cost is therefore zero. The remaining layers pay the full residual, cleanup, and light terms. There are 64531 cleanup steps and 4606 additional totient-source basepoints in these finite certificates.

For each window, put $g = 1/(6H)$ with $H$ its final height. The AP count and the same actual pool satisfy the joint response

$$|\{S \in \mathcal{F} : S \text{ is an AP}\}| + 6g \sum_{S \in \mathcal{F}} |\operatorname{Pool}(S)| \leq \frac{129}{800} N^2 + g(N-1)^3 + \frac{1}{432g^2} + NH_N.$$

The source bill is bounded using the window's upper endpoint, while this response retains the actual $N$. A cubic chord identity reduces the resulting margin to two endpoint checks. The exact rational endpoint margins are positive. The common-root proof also pays the empty-small-kernel branch and the three high-root branches, with cutoff 594 and respective parameters $(K, J) = (198, 6), (260, 5), (2083, 2)$. The assembly is `Global/N200000/CommonRoot200000.lean`.

For recovery, let $c, L, R$ be the root and endpoints of the same oriented family, let $\mathcal{E}$ be its retained omitter family, and put $w = L + N - R$. Density and the all-pair budget first give $L < N/16$ and $w < 51N/500$. Two preparations then give successively

$$|\mathcal{E}| \leq \frac{1337}{60} w - \frac{297}{20}, \qquad L < N/27, \quad w < N/15,$$

and

$$|\mathcal{E}| \leq \frac{947}{60} w - \frac{187}{20}, \qquad L < N/27, \quad w < N/16.$$

These bounds pay the native recovery gate since $2811/27 + 1193/16 < 192$. All six minimum-omitter size branches include their full fringe cost. The tightest leading coefficient has positive slack $1/240$. The exact original-family restoration is `Retention/N200000/Restore200000.lean`.

**Lemma 8** (Finite-window certificate)**.** *If* $200000 \leq N < 2^{49}$ *and* $\mathcal{F}$ *is legal, then* $|\mathcal{F}| \leq B(N)$.

*Computer-assisted proof.* The lower-end windows just described are assembled with the remaining source, common-root, and recovery windows in `Global/N200000/Global200000.lean`, declaration `actual_original_upper_bound`. Its only hypotheses are $200000 \leq N$ and legality of the original family. The arithmetic data and every required structural lemma are in its direct import closure. The files use the physical directory paths and are checked with the pinned Lean and mathlib versions as specified below. □

*Proof of Theorem 1.* For $N \geq 2^{49}$ apply Proposition 6. For $200000 \leq N < 2^{49}$, Lemma 8 gives the upper bound, and Proposition 2 gives an attaining family. The same construction proves the assertion for every positive $N$. □

## 6 Formal verification

The repository is https://github.com/lixiang90/Erdos272/tree/dev. The public entry is `formal/PublicResults.lean`. It defines the ground set, AP predicate, legality, $B(N)$, the

finite set of legal families, and its extremal value directly in that file using Lean and mathlib types. The proofs apply to these definitions by definitional equality with the internal finite-set model. In particular there is no additional common-point hypothesis in the public upper bound.

Theorem 1 is checked by `PublicResults.exact_value` and `PublicResults.paper_exact_value`; `PublicResults.upper_bound` gives the universal original-family upper bound and `PublicResults.optimal_family` supplies an attaining family, all in namespace `JSP233`. Their numerical hypothesis is $200000 \le N$. The construction entry `PublicResults.lower_bound` assumes only $1 \le N$.

The development is pinned to Lean 4.35.0 and mathlib's matching stable release. The lock file records the exact dependency revisions. From a checkout with the specified Lean toolchain, run

```
cd formal
lake exe cache get
lake build
```

The public entry prints the theorem types and their axiom dependencies. The proofs use only `propext`, `Classical.choice`, and `Quot.sound`; finite certificates are checked by Lean's kernel. A full build can require substantial time and memory. With limited resources, compile the project modules sequentially in dependency order, reuse compiled dependencies, and choose process, thread, memory, and CPU limits suited to the available hardware.

The directory groups separate foundational set geometry, private-pair counting, retention and restoration, arithmetic certificates, and global assembly. A proof covering the entire remaining finite range is still missing.

**Acknowledgements and collaboration.** The project builds on the work cited below and was developed with AI-assisted research, programming, and review. Formal checking and independent review serve different purposes: the former checks the proof terms, and the latter checks their interpretation as the stated mathematical problem. Contributions that close the remaining finite gap are welcome. Please credit the earlier results and all new contributions, and coordinate a joint application for the Justin Sun Prize.

## References

[1] T. Szabó, Intersection properties of subsets of integers, *European Journal of Combinatorics* **20** (1999), 429–444. https://page.mi.fu-berlin.de/szabo/PDF/aps.pdf.

[2] Z. Yang, Exact values and exact upper bounds for families of integers with arithmetic progression intersections (Erdős Problem #272), arXiv:2607.23004v1 (2026). https://arxiv.org/abs/2607.23004v1.

[3] J. Kohlmeyer and L. Kruer, *A linear error bound for Erdős problem 272*, manuscript and companion Lean development, fixed source commit `a624e9de77e0bd8868e7103f8b379f9dbd86b607` in JENW1N/complete-math-solutions (fixed Lean source). The source informed the structural route; the present theorems are checked in the accompanying finite-integer-set model.

[4] The Justin Sun Prize, Awards and problem list, https://github.com/TheJustinSunPrize/awards.